Chapter 2 · Story

A brighter poster. More visitors. A cause?

An original story at the Common Ground repair café.



The poster that “worked” · c02s01

A fictional story. A real question about people.

What would you do?

Which statement fits Leo’s evidence?

A The poster caused the increase.

B The poster had no effect.

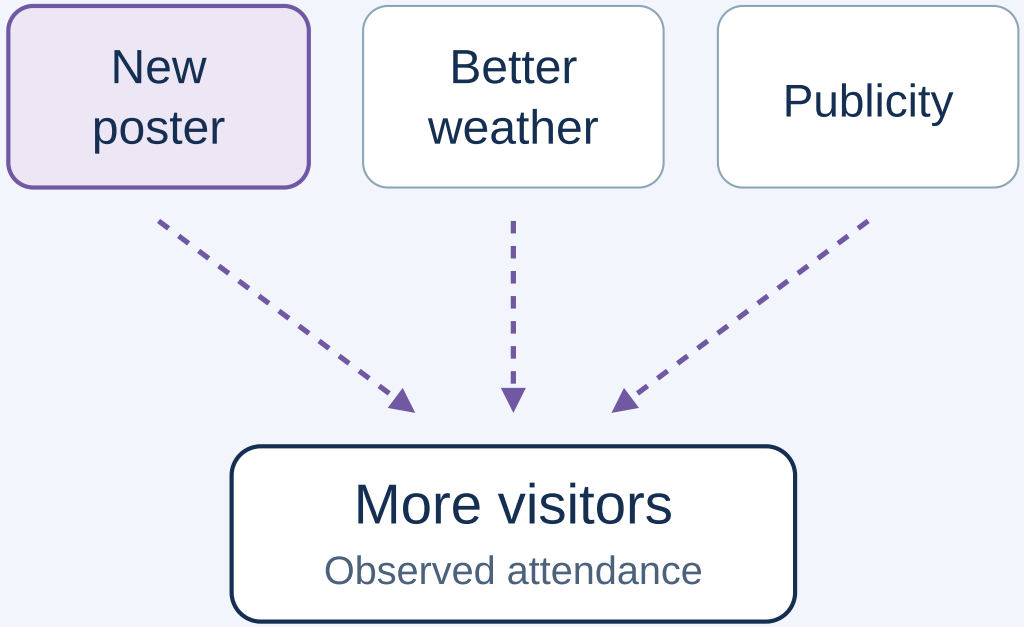
C Attendance rose; several causes remain possible.

Suggested answer: C. Saying that attendance rose preserves the observation. Leaving the cause open reflects the limits of the comparison.

Comparison makes a claim testable.

- **Correlation**: things vary together.
- **Confound**: another factor could explain the pattern.
- **Random assignment**: allocate conditions by chance.
- **Random sampling**: select people from a population by chance.

Several changes, one outcome



Dashed paths: possible explanations

A fair comparison separates explanations.

Several changes can fit the same before-and-after pattern.

Ask how much—and how uncertain.

- **Size**: 12% versus 15% is a difference of three percentage points.
- **Uncertainty**: another sample may give a different estimate.
- **Statistical significance**: judged against a model, often assuming no group difference.
- Significance does not establish practical importance; nonsignificance does not prove zero effect.

Invented attendance example

Invitation A



12 of 100 households attend

Invitation B



15 of 100 households attend

15% − 12% = 3 percentage points

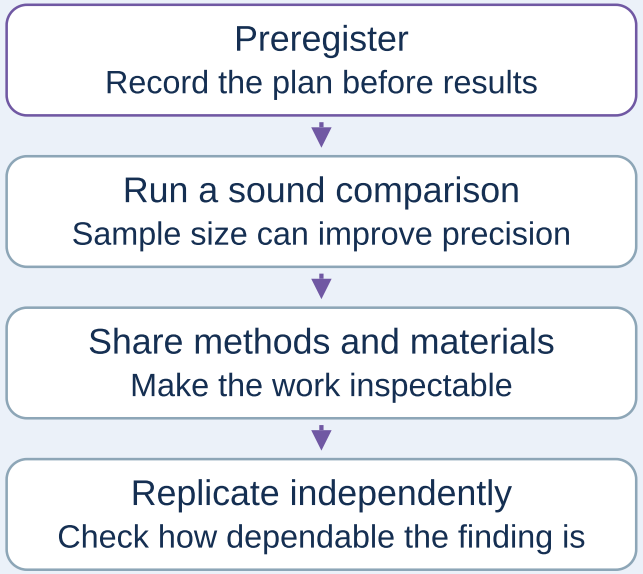
A different sample could give a different estimate.

Invented counts illustrate the size of a difference.

What can we really conclude?

Replication repeats a test to examine how dependable a finding is. Independent tests help us learn whether a result holds up elsewhere. Preregistration records the planned methods and analysis before results are known. That advance record reduces some opportunities for selective reporting. Preregistration cannot rescue a poor design. With a sound design, a larger sample can narrow the uncertainty around an estimate. More observations cannot fix a systematically unfair comparison. Sharing the study's methods and materials lets others inspect how it was done. A memorable study is a starting point for evaluation, rather than a final verdict.

Build confidence through checks



A larger sample cannot fix a biased comparison.

Plan, inspect, and independently repeat a sound comparison.

Try the idea somewhere new.

Which invitation comparison is strongest?

A Design A in winter; design B in summer.

B Randomize designs for one event; define attendance in advance.

C Design A for existing fans; design B for newcomers.

Suggested answer: B. Randomizing invitations for one event improves the comparison. Also check whether households share invitations.

Close your notes. Explain it.

> How does random assignment differ from random sampling?

> Why does the rise in attendance after Leo changed the poster fail to establish that the poster caused the increase?

1. Random assignment allocates people to conditions. Random sampling selects people from a population.

2. Other conditions can change over time and produce the same before-and-after pattern.



Maya, Sam and Leo at the repair café.

Carry one idea into your week.

Find an everyday causal claim. Name its outcome, one competing explanation, and a feasible comparison. State whose behavior the result would represent.

Revisit tomorrow and in a week.
Explain it without notes, then use a different example.



Take the question into a different setting.